DEALER TRAINING · NOVEMBER 2026

Tornado



A foliar biostimulant: amino acids 10%, fulvic acid 4.5%, humic acid 3.4% and seaweed extract 0.5%, all weight per volume (w/v), in 0.5 L, made in Lahore.

VAN wrote this deck for its dealers and field team. A dealer can repeat its words to a grower.

THE 30-SECOND EXPLANATION

What a dealer says to a grower in 30 seconds.

Tornado is a foliar biostimulant in a 0.5 L bottle. It holds amino acids at 10% w/v (100 g a litre), with fulvic acid 4.5% w/v, humic acid 3.4% w/v and seaweed extract 0.5% w/v. It is not a nutrient source in the crop plan. A plant normally builds its amino acids from nitrate, and this uses energy. In a hot week, a stressed crop closes its leaf pores, makes less sugar and has less energy to spare. Sprayed amino acids are already made, so the plant does not spend energy making them. Fulvic acid carries nutrients into the leaf, humic acid helps the droplet stay on it, and seaweed extract acts on the plant's signals. Per acre, wheat takes 2 bottles; basmati rice, watermelon, citrus and strawberry take 1. Spray it in 100 to 200 L of water an acre, never at midday. If the crop is short of a nutrient, correct that first. VAN gives no recovery-time or yield figure.

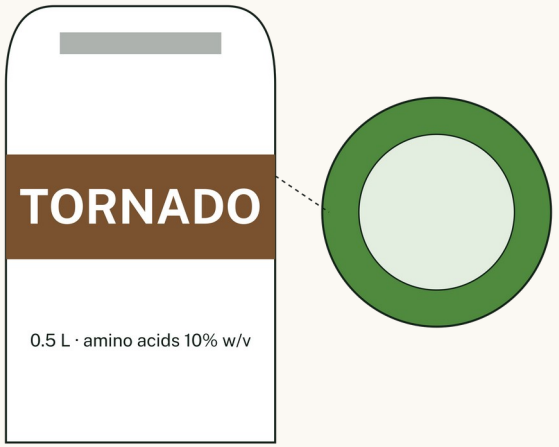
The 4 points it must cover

1. The bottle: amino acids 10% w/v, 100 g a litre, with fulvic acid 4.5%, humic acid 3.4% and seaweed extract 0.5%, all w/v. 0.5 L, PS 5606-2023. 50 g of amino acids in a bottle.
2. The mechanism: ready-made amino acids; fulvic acid carries nutrients into the leaf; humic acid holds the droplet on it; seaweed extract acts on plant signals.
3. The stage and method: grand growth on wheat, watermelon and citrus, early growth on basmati rice and strawberry. Foliar, water first, morning or evening.
4. The limits: it is not a nutrient source in the crop plan. If the crop is short of nitrogen, potash or a micronutrient, correct that. No figure for how fast the crop recovers, no yield figure, no disease claim.

Explain the product in simple words a grower uses, and keep to the figures on these slides. Every figure in this 30-second explanation is on the bottle, in the published crop plan or in the named evidence on the slides that follow.

THE BOTTLE

What the bottle says, and what it is registered as.



0.5 L

the bottle. It holds 50 g of amino acids at the declared 10% w/v. Per acre, wheat takes 2 bottles; basmati rice, watermelon, citrus and strawberry take 1.

10% w/v

amino acids, 100 g in a litre, declared on the label. VAN declares them as free amino acids, in the L form the plant's enzymes use, extracted from plant sources.

3 more

fulvic acid 4.5%, humic acid 3.4% and seaweed extract 0.5%, all w/v, as the label declares them: 22.5 g, 17 g and 2.5 g in the bottle.

Foliar

sprayed on the leaf, in 100 to 200 L of water an acre. Do not recommend it for the soil or the drip.

795

rupees a bottle, delivered farmer price (VAN farmer price sheet, August 2026). The 2 bottles an acre on wheat cost Rs 1,590.

LAB 336

every batch is released through VAN Lab, accredited by the Pakistan National Accreditation Council (PNAC), before it ships.

Registered as	Pakistan Standard	Manufacturing licence
Tornado	PS 5606-2023	CM/L-4125/2025

If a grower asks about registration, give the licence number. Do not give an expiry date for the licence.

A GROWER'S QUESTION

“It touched 45 degrees in June and my crop just stood still for 2 weeks. The soil had everything. What happened to it, and is there anything for that week on summer crops?”

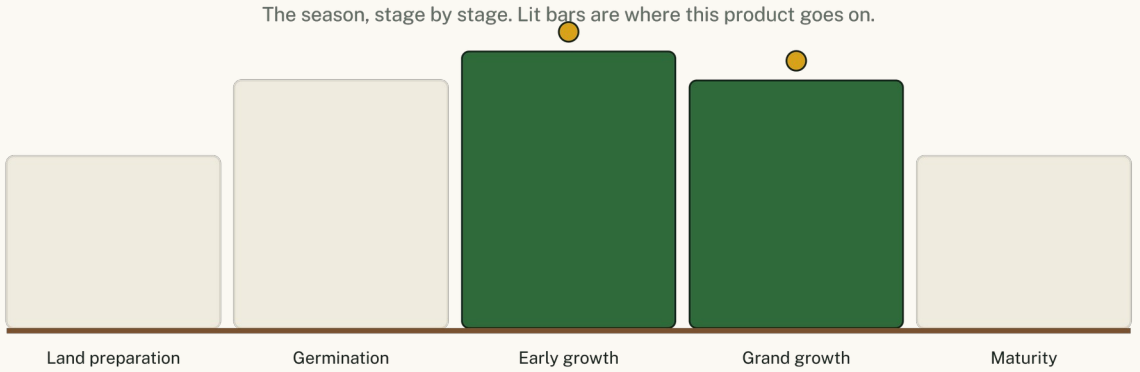
The grower says his soil had everything the crop needed. The heat reduced the plant’s ability to grow. The answer has 3 parts: what a plant does under heat stress, what the 4 parts of the formulation do about it, and what VAN will not promise.

The next 3 slides give the 3 answers, 1 idea on each slide.

ANSWER 1 OF 3

Under heat stress, the plant makes less sugar and uses more energy.

In a hot week the leaf closes its pores to save water. With the pores shut it takes in little carbon dioxide and makes little sugar, so its supply of energy falls. At the same time, the plant uses more energy. It builds its own protective compounds and amino acids, and making an amino acid from nitrate costs energy, because the plant must first convert the nitrate chemically. If nothing is short in the soil, what is short is the plant's energy to keep working through the hot period, and a biostimulant is used in that situation. A biostimulant does not replace a nutrient: if the crop is short of a nutrient, correct that first.



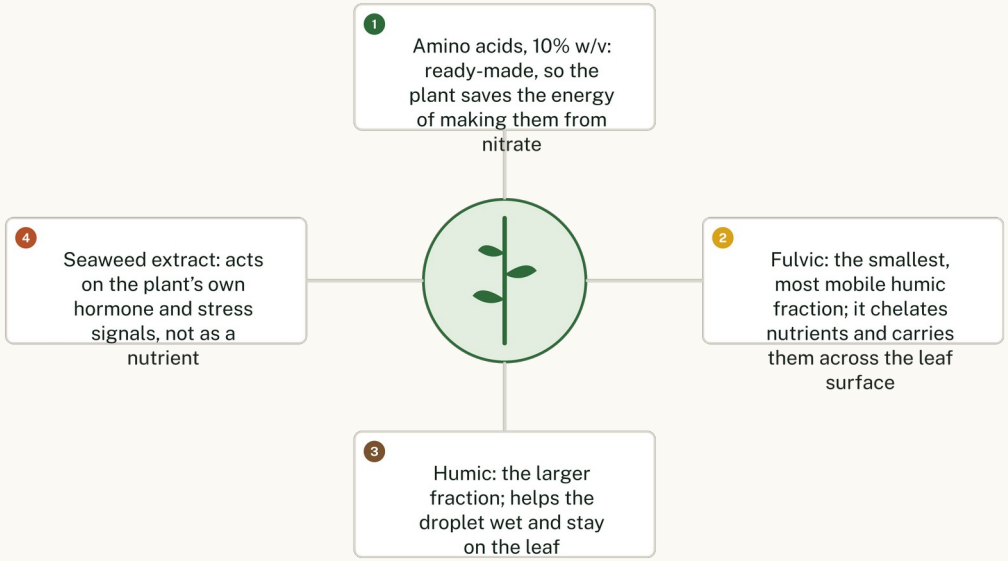
The stage bar shows where the crop plan puts Tornado: grand growth on wheat, watermelon and citrus, early growth on basmati rice and strawberry. The drawing illustrates the idea; it is not a measurement.

Say it like this: "Your crop was not short of nutrients. The heat made the plant make less sugar and use more energy. A bag of fertiliser does not solve that problem."

ANSWER 2 OF 3

The 4 parts of the formulation each do a different job.

Amino acids, declared at 10% w/v, are the building blocks of protein. Sprayed on the leaf, they are already made, so the crop does not spend energy making them from nitrate. Fulvic acid is the smallest and most mobile part of the humic material. It binds positively charged nutrients and carries them into the leaf, so it suits a product sprayed on the leaf. Humic acid is the larger fraction; it helps the droplet wet the leaf and stay on it. Seaweed extracts are used in agronomy for their effect on the plant's own hormone and stress signalling, not for their nutrient content. Present the product as 4 ingredients with 4 different jobs in 1 spray, not as a fertiliser.



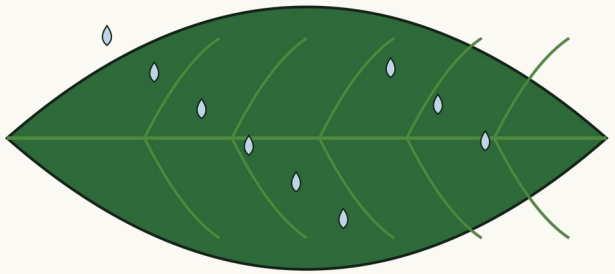
The 4 parts, from textbook biostimulant chemistry. Declared: amino acids 10%, fulvic acid 4.5%, humic acid 3.4% and seaweed extract 0.5%, all w/v.

Say it like this: "It gives the plant amino acids it does not have to make, fulvic acid that carries nutrients into the leaf, humic acid that keeps the droplet on the leaf, and seaweed that acts on the plant's signals."

ANSWER 3 OF 3

Spray it on the leaf at the stage the crop plan gives; VAN promises no result figure.

Tornado goes on the leaf, in 100 to 200 L of water an acre, as a fine, even fan, in the morning or the evening. A droplet that dries in seconds at midday has not been absorbed. Wheat: 2 bottles at grand growth. Basmati rice: 1 bottle at early growth. Watermelon: 1 bottle at grand growth. Citrus: 1 bottle at grand growth. Strawberry: 1 bottle at early growth. VAN promises no number: no days recovered, no yield gained, and no temperature it protects against. No trial of Tornado is on VAN's record, so this training gives only the mechanism and the crop plan rows.



Sprayed onto the leaf. Small doses, taken up through the leaf surface, for a nutrient the crop needs quickly or that the soil locks up. Spray early morning or late afternoon; never on a wilted crop.

The leaf with the spray on it. No absorption percentage or timing figure is given.

Say it like this: "Put the water in first, then the bottle, and spray in the morning or evening at the stage your plan gives. I will not tell you it saves X days, because nobody has measured that."

THE EVIDENCE

What is verified for Tornado, and what is not.

No trial of Tornado itself is on VAN's record. A dealer may state the following as facts, and should say where each comes from: what the label declares, the licence, VAN's declaration on the amino acids and the published chemistry below.

Item	What it says	How a dealer may use it
The label	Tornado, amino acids 10% w/v (100 g/L), fulvic acid 4.5% w/v, humic acid 3.4% w/v and seaweed extract 0.5% w/v. 0.5 L.	As fact: the 4 declared figures.
The licence	Pakistan Standard PS 5606-2023 (the biostimulant class), manufacturing licence CM/L-4125/2025.	As fact. The number, never an expiry date.
VAN's declaration	The amino acids are free, in the L form, extracted from plant sources, as the certificate of analysis supports (VAN, 21 August 2026).	As VAN's statement about its own product. Never as a comparison with a named brand.
The chemistry (published)	Nitrate reduction inside the plant costs energy; amino acids supplied to the leaf do not need that step. Only free L-form amino acids are directly usable; harsh acid processing (hydrolysis) turns L into a mixture of L and D and destroys some amino acids. Fulvic and humic fractions bind positively charged nutrients (cations) and carry them.	As general plant physiology and humic chemistry, spoken. Never as a rate or a result for this product.
The soil, for context	Mean pH 8.22 across 770,160 Punjab soil survey samples (about 2016 to 2018); 69.1% above pH 8. A foliar spray places a chelated micronutrient on the leaf, where calcareous soil cannot lock it up.	As the reason a foliar route exists. Never as a Tornado result.
Not on VAN's record	A Tornado trial; a yield, recovery-time or stress-tolerance figure; a disease effect; a tank-mix compatibility list; a 1 L pack or a drone rate.	Not said.

Sources: the Tornado label and VAN's certificate of analysis (21 August 2026); the regulatory table on van.com.pk; the Punjab soil survey (770,160 samples, tested in about 2016 to 2018); standard plant physiology and humic-substance chemistry.

On wheat, the grower sprays 100 g of amino acids on the leaf with 2 bottles of Tornado, at Rs 1,590 per acre.

The price is the delivered farmer price from the VAN farmer price sheet, August 2026. Amino acids are from the declared 10% w/v. Per acre, wheat takes 2 bottles; basmati rice, watermelon, citrus and strawberry take 1.

Tornado, amino acids 10% w/v	Volume	Amino acids	Water on the acre	Rupees
1 bottle of 0.5 L	0.5 L	50 g		795
Basmati rice, watermelon, citrus and strawberry: 1 bottle an acre	**0.5 L**	**50 g**	**100 to 200 L**	**795**
Wheat: 2 bottles an acre	**1 L**	**100 g**	**100 to 200 L**	**1,590**

On wheat, the grower spends Rs 1,590 per acre to spray 100 g of amino acids, with the fulvic acid, humic acid and seaweed extract, on the leaf at grand growth, in 100 to 200 L of water. The water volume is part of the rate: if the water is halved, the spray is 2 times as concentrated. VAN publishes no yield or crop-response figure for Tornado.

Price: the VAN farmer price sheet, August 2026. 50 g = 0.5 L × 100 g/L. Rates: VAN's published crop plans on van.com.pk (wheat 2 bottles at grand growth; basmati rice 1 bottle at early growth; watermelon 1 bottle at grand growth; citrus 1 bottle at grand growth; strawberry 1 bottle at early growth). Water: VAN's spray guidance.

The crop plans use 1 or 2 bottles of Tornado per acre, at Rs 795 or Rs 1,590.

Bottles per acre are from the published crop plans; rupees per acre are calculated from the delivered farmer price in the VAN farmer price sheet, August 2026. 5 crop plans name Tornado.

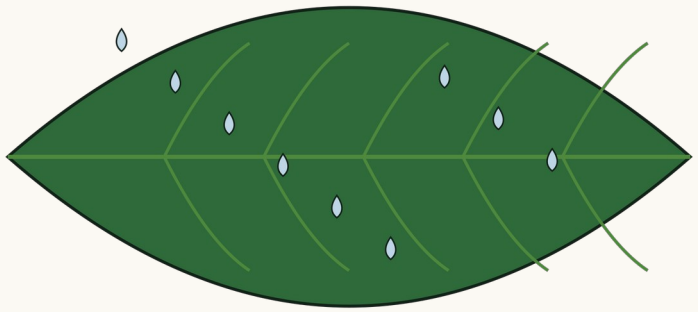
Crop	Tornado per acre	Amino acids	Rupees per acre	Stage and method, as printed
Wheat	2 bottles of 0.5 L	100 g	1,590	Grand Growth, spray
Basmati rice	1 bottle of 0.5 L	50 g	795	Early Growth, spray
Watermelon	1 bottle of 0.5 L	50 g	795	Grand Growth, spray
Citrus	1 bottle of 0.5 L	50 g	795	Grand Growth, spray
Strawberry	1 bottle of 0.5 L	50 g	795	Early Growth, spray

Grand growth is the main stage in the crop plans: the crop is under most stress then, and the size of the harvest is being set. Basmati rice and strawberry take it at early growth, the stage when those crops are most exposed to stress. For a crop outside these 5 there is no printed rate, and a dealer says so.

Rows as printed in VAN's published crop plans on van.com.pk; 0.5 L bottle at Rs 795. Rupees = bottles × 795.

HOW IT IS APPLIED

On the leaf: water first, then the bottle, in the morning or evening, never at midday.



Sprayed onto the leaf. Small doses, taken up through the leaf surface, for a nutrient the crop needs quickly or that the soil locks up. Spray early morning or late afternoon; never on a wilted crop.

Water first, product second

Fill the tank with 100 to 200 L of clean water for the acre, then add the bottles the crop plan gives for the acre and keep it agitated. If the tank is not kept agitated, the product settles, and the end of the spray gets a different dose from the start. Never add the product before the water.

A fine, even fan, when the leaf can absorb it

Wet the leaf without letting the spray run off. Absorption happens while the droplet is still wet; a droplet that dries in seconds at midday in June has been wasted. Spray in the morning or the evening.

At the stage your plan gives; jar-test any mix

Grand growth on wheat, watermelon and citrus, early growth on basmati rice and strawberry. Tornado mixes in water; keep the tank agitated so it does not settle. A biostimulant, a micronutrient and a pesticide may not mix well in 1 tank: jar-test them first. Not for the soil or the drip. Tornado is approved for drone spraying, but no drone rate is published: refer any drone question to VAN.

OBJECTION 1 OF 8

“795 for 0.5 L? What is actually in it?”

What the dealer says

Write down the price, and then what is declared: 50 g of amino acids in the bottle, free, in the L form the plant uses, from plant sources, with 22.5 g of fulvic acid, 17 g of humic acid and 2.5 g of seaweed extract in the same bottle. Per acre, wheat takes 2 bottles; basmati rice, watermelon, citrus and strawberry take 1; the 2 bottles on wheat cost Rs 1,590. It is sprayed once, at the stage the crop plan gives, when the crop is under most stress.

The facts behind the answer

- 0.5 L, Rs 795, delivered; wheat 2 bottles, Rs 1,590 (the farmer price sheet, August 2026; the crop plans).
- 50 g of amino acids per bottle: 0.5 L × 100 g/L (the declared 10% w/v).
- Fulvic acid 4.5%, humic acid 3.4% and seaweed extract 0.5%, all w/v, as the label declares them: 22.5 g, 17 g and 2.5 g per bottle.
- Free, L-form, plant-sourced amino acids (VAN’s certificate of analysis, 21 August 2026).

Never say: “It gives you X more maunds” or “it saves the crop”. No yield or recovery-time figure exists for the product.

OBJECTION 2 OF 8

“It has amino acids. Is that nitrogen? Can I cut some urea?”

What the dealer says

No. 100 g of amino acids on an acre of wheat is not a nitrogen dose, and VAN does not sell it as one. A bag of urea supplies kilograms of nitrogen. Tornado supplies grams of amino acids, which the plant does not have to make itself. Sprayed on the leaf, they are already made, so the crop does not spend energy making them from nitrate. Keep the nitrogen in your crop plan as printed.

The facts behind the answer

- 50 g of amino acids per bottle; 100 g in the 2 bottles on wheat (the declared 10% w/v).
- The nitrogen line on the wheat crop plan is a separate set of rows (the crop plan on van.com.pk).
- A biostimulant does not add nitrogen. Amino acids sprayed on the leaf are already made, so the crop does not spend energy making them from nitrate (standard plant physiology).

Never say: “It replaces a top dressing” or any nitrogen equivalent for the amino acids.

OBJECTION 3 OF 8

“My crop is already scorched from the heat. Will this bring it back?”

What the dealer says

I will not promise that, and VAN does not. Nobody has measured what Tornado does to a scorched crop, so there is no figure and no rescue claim. The crop plan gives it at grand growth or early growth, depending on the crop, when the crop is under most stress. The leaf absorbs it while the droplet is still wet. If the crop is short of a nutrient, correct that first. Use it at the stage your plan gives, and judge it in your own field.

The facts behind the answer

- No trial of Tornado is on VAN’s record, so no recovery-time or yield figure is printed.
- Grand growth on wheat, watermelon and citrus, early growth on basmati rice and strawberry (the crop plans on van.com.pk).
- If the crop is short of a nutrient, the nutrient comes first; a biostimulant does not replace it.

Never say: “It recovers the crop in X days”, “it protects against 45 degrees” or “it undoes heat damage”.

OBJECTION 4 OF 8

“Can I mix it with my zinc spray and my pesticide?”

What the dealer says

Tornado mixes in water; keep the tank agitated so it does not settle. Its fulvic acid carries nutrients into the leaf. But a biostimulant, a micronutrient and a pesticide are 3 different kinds of chemical and may not mix well, and VAN publishes no compatibility list. Test a jar first: mix the 3 in the same proportions in a jar, wait, and look for settling or curdling. If the jar stays clear, spray. If not, spray them separately.

The facts behind the answer

- Jar-test any new tank mix before spraying an acre (VAN’s spray guidance).
- Fulvic acid chelates cations and carries them across the leaf surface (established humic-substance chemistry).
- VAN publishes no tank-mix compatibility list for Tornado.

Never say: “It is compatible with everything” or that it “improves” a named pesticide.

OBJECTION 5 OF 8

“How much water, and what time of day?”

What the dealer says

Use 100 to 200 L of clean water for the acre: water first, then the bottles the crop plan gives, kept agitated. Spray a fine, even fan that covers the leaf without running off. Spray in the morning or evening, never at midday: the leaf absorbs it while the droplet is wet, and a droplet that dries in seconds on a June afternoon is wasted. The water volume is part of the rate: if the water is halved, the spray is 2 times as concentrated.

The facts behind the answer

- 100 to 200 L of water an acre, water first, morning or evening, a fine fan (VAN’s spray guidance).
- Per acre, wheat takes 2 bottles; basmati rice, watermelon, citrus and strawberry take 1 (the crop plans on van.com.pk).
- Absorption happens while the droplet is wet; do not spray at midday in a Punjab summer.

Never say or give: any drone or ultra-low-volume rate. None is printed for Tornado.

OBJECTION 6 OF 8

“Which stage, and for which crops?”

What the dealer says

Wheat: 2 bottles at grand growth. Basmati rice: 1 bottle at early growth. Watermelon: 1 bottle at grand growth. Citrus: 1 bottle at grand growth. Strawberry: 1 bottle at early growth. Those are the 5 crop plans that name it. For any other crop there is no printed rate, and a dealer says so instead of guessing; ask VAN.

The facts behind the answer

- 5 rows across 5 crop plans: spray on every row; the 0.5 L pack on every row (the crop plans on van.com.pk).
- No rate is published for any crop outside the 5.

Never say or give: a rate for cotton, maize, potato or any crop outside the 5.

OBJECTION 7 OF 8

“Is it registered? What is the licence?”

What the dealer says

It is registered under Pakistan Standard PS 5606-2023, the biostimulant class, with manufacturing licence CM/L-4125/2025, and VAN is the registered maker. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Enter the batch number from the bottle on van.com.pk, and you can read the certificate for that batch.

The facts behind the answer

- Tornado: PS 5606-2023, CM/L-4125/2025 (the regulatory table on van.com.pk).
- PS 5606-2023 is the biostimulant product class; the VL liquids sit under PS 5469-2020, a different class.
- Every batch is released through VAN Lab, PNAC LAB 336, ISO/IEC 17025:2017.

Never say or give: a licence expiry date, or the licence as a claim about performance.

OBJECTION 8 OF 8

“Another bottle says 20% amino acids. Yours says 10%.”

What the dealer says

Do not compare bottles on the total amino acid figure. A total figure can include bound protein the plant cannot take up; only free amino acids are taken up directly. Harsh acid processing turns the L form the plant’s enzymes use into a mixture of L and D, and destroys some amino acids. The source matters too. VAN declares Tornado’s amino acids as free, L-form and plant-sourced, and the certificate supports it. Ask 3 questions about any bottle: are the amino acids free or total, are they L-form or mixed, and what is their source? Then compare like with like.

The facts behind the answer

- Only free amino acids are taken up directly; a total figure can include bound protein (standard biostimulant chemistry).
- Harsh acid hydrolysis turns L-amino acids into an L and D mixture and destroys some of them (standard biostimulant chemistry).
- Tornado’s amino acids are declared free, L-form and plant-sourced, supported by the certificate of analysis (VAN, 21 August 2026).
- Declared at 10% w/v on the label.

Never say or give: a competitor’s brand name, or “theirs is D-form” about any named product. The 3 questions are asked generally.

ITS PLACE IN THE CROP PLANS

Tornado in the published crop plans, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Grand Growth	2 packs	Spray	0.5 L
Basmati Rice	Early Growth	1 pack	Spray	0.5 L
Watermelon	Grand Growth	1 pack	Spray	0.5 L
Citrus, 5 years and over	Grand Growth	1 pack	Spray	0.5 L
Strawberry	Early Growth	1 pack	Spray	0.5 L

The rows are printed as published in the crop plans on van.com.pk. Coverage: these are all 5 rows in the 5 crop plans that name Tornado. Every row says spray and uses the 0.5 L pack. Each rate is per acre, and the crop page scales it to the size of the field.

WHAT VAN DOES NOT CLAIM FOR THIS PRODUCT

A dealer who makes any of these claims is speaking for himself, not for VAN.

1. A yield figure, a days-recovered figure or a temperature it protects against. No Tornado trial exists; what it does in a hot week is explained from plant physiology only.
2. Any figure for the fulvic acid, humic acid or seaweed extract other than the declared 4.5%, 3.4% and 0.5% w/v.
3. That it adds nitrogen, replaces any nutrient in the crop plan, or corrects a nutrient shortage. If the crop is short of a nutrient, correct that first.
4. A disease, immunity or frost effect. No such effect has been measured, and the rules for the biostimulant class do not allow the claim.
5. A tank-mix compatibility guarantee. Jar-test every new mix; VAN publishes no list.
6. A 1 L bottle, a drone rate, or a rate for any crop outside the 5 crop plans that name it.

HOW A GROWER VERIFIES A BOTTLE

Every batch is tested, and the batch number on the bottle identifies its certificate.

1 · The batch number

Every bottle carries the batch number. Read it to the grower and show him where it is printed.

2 · The certificate of analysis

On van.com.pk, a grower can read the certificate of analysis for any bottle by entering the batch number printed on it. The grower does not need to post a sample.

3 · The laboratory

VAN Lab is accredited by the Pakistan National Accreditation Council (PNAC) to ISO/IEC 17025:2017, the international standard for testing laboratories, as LAB 336. Every batch is released through the laboratory before it ships. The certificate carries the laboratory's number.

What a certificate for Tornado shows

The certificate shows the tested amino acid content of the batch against the declared 10% w/v. It also shows the batch number, and the Pakistan Standard PS 5606-2023 and licence CM/L-4125/2025 printed on the bottle. VAN states that the certificate confirms the amino acids are free, L-form and from plant sources. This training shows no batch result; a grower can read the certificate of his own batch on van.com.pk by its batch number.

THE URDU LINES · REVIEW STATUS UNDER EACH LINE

The lines a dealer says in Urdu.

The first line is the product's main message. Where its nutrient has one, the line gives VAN's promise word: the 1-word brand message VAN uses for that nutrient. The other 3 lines are key sentences of this deck. Each English line gives the meaning of the Urdu beside it. Send any correction to partner@van.com.pk.

Tornado is not a fertiliser. Its amino acids are ready-made, so a crop under heat stress does not spend energy making them. *English meaning · Urdu line pending review*	یہ فصل کو خوراک نہیں دیتا، مشکل ہفتے میں فصل کا خرچ کم کرتا ہے
Water first, then the bottle. Morning or evening, never midday. *English meaning · Urdu reviewed 27 Sep 2026*	پہلے پانی، پھر بوتل۔ صبح یا شام، دوپہر میں کبھی نہیں
If a nutrient is short, fix that first. This is not a substitute for a bag. *English meaning · Urdu reviewed 27 Sep 2026*	اگر کسی غذائی جز کی کمی ہے تو پہلے وہ پوری کریں، یہ بوری کا بدل نہیں
Sprayed at the stage the plan prints. No promise of days or maunds. *English meaning · Urdu reviewed 27 Sep 2026*	اسی مرحلے پر چھڑکیں جو پلان میں لکھا ہے۔ نہ دنوں کا وعدہ، نہ من کا

TALK TO VAN

Use the address that matches your request.

For growers

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch or a dealer near you.

For the trade

partner@van.com.pk

Dealers, distributors and the trade: stock, training and corrections to this deck.

For everything else

info@van.com.pk

Supply, the laboratory or a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore, has 1 helpline and 3 addresses. VAN sends no group messages; every grower who writes gets a direct reply.

Tornado is sprayed on the leaf, water first, morning or evening, at the stage the plan gives; VAN gives no result figure.

